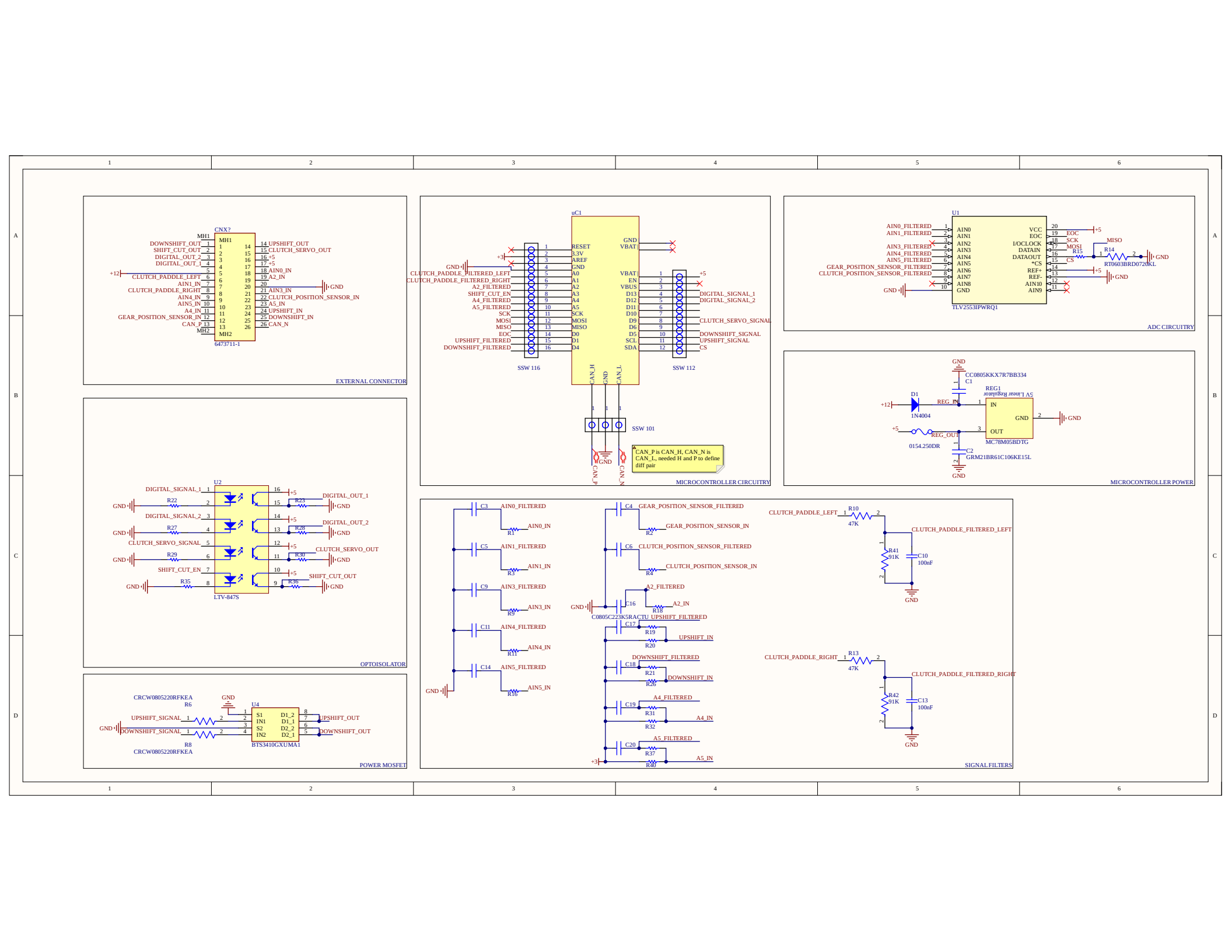

CNX?
MH1
DOWNSHIFT_OUT
SHIFT_CUT_OUT
DIGITAL_OUT_2
DIGITAL_OUT_1
+12
CLUTCH_PADDLE_LEFT
AIN1_IN
CLUTCH_PADDLE_RIGHT
AIN4_IN
AIN5_IN
A4_IN
GEAR_POSITION_SENSOR_IN
CAN_P
MH2
UPSHIFT_OUT
CLUTCH_SERVO_OUT
+5
A0_IN
A2_IN
GND
AIN3_IN
CLUTCH_POSITION_SENSOR_IN
A5_IN
UPSHIFT_IN
DOWNSHIFT_IN
CAN_N
6473711-1
EXTERNAL CONNECTOR
uC1
RESET
3.3V
AREF
A0
A1
A2
A3
A4
A5
SCK
MOSI
MISO
D0
D1
D4
VBAT
EN
VBUS
D13
D12
D11
D10
D9
D6
D5
SCL
SDA
CLUTCH_PADDLE_FILTERED_LEFT
CLUTCH_PADDLE_FILTERED_RIGHT
A2_FILTERED
SHIFT_CUT_EN
A4_FILTERED
A5_FILTERED
UPSHIFT_FILTERED
DOWNSHIFT_FILTERED
DIGITAL_SIGNAL_1
DIGITAL_SIGNAL_2
CLUTCH_SERVO_SIGNAL
DOWNSHIFT_SIGNAL
UPSHIFT_SIGNAL
CS
SSW 116
SSW 112
SSW 101
CAN_H
CAN_L
CAN_P is CAN_H, CAN_N is CAN_L, needed H and P to define diff pair
MICROCONTROLLER CIRCUITRY
U1
AIN0_FILTERED
AIN1_FILTERED
AIN3_FILTERED
AIN4_FILTERED
AIN5_FILTERED
GEAR_POSITION_SENSOR_FILTERED
CLUTCH_POSITION_SENSOR_FILTERED
AIN0
AIN1
AIN2
AIN3
AIN4
AIN5
AIN6
AIN7
AIN8
VCC
EOC
I/OCLOCK
DATAIN
DATAOUT
*CS
REF+
REF-
AIN10
AIN9
MISO
MOSI
SCK
CS
R14
R15
RT0603BRD0720KL
TLV2553IPWRQ1
ADC CIRCUITRY
CC0805KKX7R7BB334
C1
REG1
5V Linear Regulator
D1
1N4004
REG_IN
REG_OUT
0154.250DR
IN
OUT
MC78M05BDTG
C2
GRM21BR61C106KE15L
MICROCONTROLLER POWER
U2
DIGITAL_SIGNAL_1
DIGITAL_SIGNAL_2
CLUTCH_SERVO_SIGNAL
SHIFT_CUT_EN
R22
R27
R29
R35
R23
R28
R30
R36
DIGITAL_OUT_1
DIGITAL_OUT_2
CLUTCH_SERVO_OUT
SHIFT_CUT_OUT
LTV-847S
OPTOISOLATOR
CRCW0805220RFKEA
R6
R8
U4
S1
IN1
S2
IN2
D1_2
D1_1
D2_2
D2_1
BTS3410GXUMA1
POWER MOSFET
C3
C5
C9
C11
C14
AIN0_FILTERED
AIN1_FILTERED
AIN3_FILTERED
AIN4_FILTERED
AIN5_FILTERED
R1
R3
R9
R11
R16
AIN0_IN
AIN1_IN
AIN3_IN
AIN4_IN
AIN5_IN
C4
C6
C16
C17
C18
C19
C20
GEAR_POSITION_SENSOR_FILTERED
CLUTCH_POSITION_SENSOR_FILTERED
A2_FILTERED
DOWNSHIFT_FILTERED
A4_FILTERED
A5_FILTERED
R2
R4
R18
R19
R20
R21
R31
R32
R37
R40
GEAR_POSITION_SENSOR_IN
CLUTCH_POSITION_SENSOR_IN
A2_IN
UPSHIFT_IN
DOWNSHIFT_IN
A4_IN
A5_IN
C0805C223K5RACTU
UPSHIFT_FILTERED
CLUTCH_PADDLE_LEFT
CLUTCH_PADDLE_RIGHT
R10
R13
47K
R41
R42
91K
C10
C13
100nF
CLUTCH_PADDLE_FILTERED_LEFT
CLUTCH_PADDLE_FILTERED_RIGHT
SIGNAL FILTERS